YOLOv8 and FaceNet Algorithm for Real-Time Face Recognition Attendance System

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***Abstract*—Several studies have explored the integration of facial recognition technologies with attendance systems to address the issues associated with manual attendance systems, such as imperfections and fraud, including attendance substitution. Despite these advances, a comprehensive evaluation comparing the performance metrics of these systems remains lacking, particularly in terms of detection speed and recognition accuracy. This study aims to fill this gap by developing a real-time facial recognition attendance system that leverages the You Only Look Once version 8 (YOLOv8) and FaceNet algorithms. The study begins with data and literature collection, followed by preprocessing of the LFW dataset into a suitable format. Next, two methods are implemented: YOLOv8-FaceNet and Dlib-ResNet, which are used to detect and recognize faces, respectively. The system's performance is evaluated based on accuracy and processing speed, providing insights into its feasibility in real-world scenarios. The test results indicate that the Equal Error Rate (EER) of the YOLOv8+FaceNet model is 0.013, with False Acceptance Rate (FAR) and False Rejection Rate (FRR) of 0.010 and 0.016, respectively. These values are at a threshold of 0.4273. Then, the EER of the Dlib+ResNet model is 0.0010, with FAR and FRR of 0.008 and 0.012. The optimum threshold of the model is 0.3909. Performance testing of both models with their respective optimum thresholds reveals that the YOLOv8+FaceNet model proposed by us has a higher accuracy than Dlib-ResNet, specifically 98.7% compared to 98.6%.**

***Index Terms*—you only look once, facenet, face recognition, attendance, threshold**

I. Introduction

In the era of digital transformation, various biometric technologies have become core components in improving security and efficiency [1]. Facial recognition is particularly useful as it is non-intrusive and fast [2]. Manual attendance systems, often used in conventional settings, have inherent problems, including imperfections and fraud, such as attendance substitution [3]. Therefore, recording attendance data using manual methods is a cumbersome and unproductive task [4]. A reliable, automated, and real-time input method is a research challenge.

Recent developments in deep learning have significantly advanced facial recognition systems. The you only look once (YOLO) algorithm proposed by Redmon *et al*. [5] revolutionized object detection by treating it as a single regression problem, allowing real-time operation with adequate detection performance. Each version has added new insights to the body of knowledge, and the latest evolution, YOLOv8, is no exception, iterating on the base architecture using new architectural designs to enhance precision and speed. In parallel, FaceNet, introduced by Schroff *et al.* [6], offers a highly effective facial recognition model by projecting facial images into a compact Euclidean embedding space using a triplet loss function, allowing accurate similarity comparison between faces. The synergy between the robust detection capabilities of YOLOv8 and the discriminative facial embedding of FaceNet provides a strong and efficient solution for implementing real-time facial recognition-based attendance systems.

Many papers have discussed incorporating facial recognition technologies into attendance systems, such as an integrated face detection and recognition system (IFDRS) by Mumtaz *et al.*, which specifically uses YOLOv8 for face detection [7]. Correspondingly, Saranya *et al.* proposed an intelligent attendance monitoring solution primarily utilizing YOLOv8 and deep learning to track student participation and focus during lectures, identifying who was present and paying sufficient attention through automated facial analysis. Both efforts exemplify the harnessing of computer vision for academic attendance purposes, yielding promising results. Nevertheless, further assessment is needed to evaluate the accuracy and processing speed of such systems.

This study aims to fill this gap by developing a real-time face recognition attendance system that utilizes the YOLOv8 and FaceNet algorithms. This study begins with the collection of data and literature, followed by data preprocessing to prepare the Labeled Faces in the Wild (LFW) dataset in a format suitable for use. Next, two methods are implemented: YOLOv8-FaceNet and Dlib-ResNet, which are used to detect and recognize faces, respectively. The system's performance will be evaluated based on accuracy and processing speed, providing insights into its applicability in real-world scenarios.

To the best of our knowledge, no study has evaluated the performance of a real-time face recognition attendance system using YOLOv8 and the FaceNet algorithm. The main contributions of this study are as follows:

1) A novel IFDRS method, YOLOv8+FaceNet, which has higher accuracy than state-of-the-art methods.
2) An IFDRS method with optimum accuracy as a face recognition-based attendance system.
3) An optimum threshold determination method for FaceNet using equal error rate (EER).

The remainder of this paper follows a systematic structure: Section II presents the main reference papers that we compare in this research. Section III is the research steps and their configuration. Section IV explains the test results. Finally, Section V answers the research objectives.

TABLE I
RELATED WORKS ON FACE RECOGNITION-BASED ATTENDANCE SYSTEMS.

Cite	IFDRS	YOLO	YOLOv8	FaceNet	Attendance System
[11]	✓	✓	✗	✓	✗
[12]	✓	✓	✗	✓	✗
[13]	✗	✗	✓	✗	✓
[14]	✗	✗	✓	✗	✓
[15]	✓	✗	✗	✓	✓
[16]	✓	✗	✗	✓	✓
Proposed Method	✓	✓	✓	✓	✓

II. RELATED WORKS

Some of our previous studies have discussed advances in smart classrooms, including attendance systems. In our earlier work, we proposed an attendance system that prevents fraud using RFID cards [8]. Our solution is to implement Shamir's secret share so that when students want to tap, the tapping must be verified with a lecturer. Additionally, we have conducted several studies on dynamic device pairing for IoT devices in smart classrooms [9]. This is to maintain user comfort because, without dynamic device pairing, entry-level users will have difficulty using electrical tools in a smart classroom [10]. Table I shows a summary of research gaps from previous research on face recognition-based attendance systems.

Several studies have combined YOLO and FaceNet for the purpose of IFDRS. Pham *et al.* [11] used YOLO for face detection and FaceNet for face recognition. However, the study has not used YOLOv8. Prayogo *et al.* [12] used YOLOv5 and FaceNet for face detection and recognition. The case study of this research focuses on an authentication system within an electronic-based government system. This presents an opportunity to develop attendance systems using YOLOv8 and FaceNet.

Several studies have applied YOLOv8 as an attendance system. Hasaraddi *et al.* [13] applied YOLOv8 to assess class attendance at a school. The system's accuracy was 0.95, obtained from training data comprising 50 students. Lam *et al.* [14] used YOLOv8 to not only detect students for attendance but also detect other attributes such as shirts, pants, and ID cards. The system is to check students' clothing.

Several studies have utilized FaceNet for attendance systems that rely on face recognition. Pham *et al.* [15] used FaceNet together with a multi-task cascaded convolutional neural network (MTCNN) for face detection and FaceNet for face recognition for student attendance. The study obtained satisfactory results under normal lighting conditions and low failure rates. Kuang *et al.* [16] used Haar-cascade for face detection and FaceNet for face recognition. However, the study was not applied to virtual class attendance. There is a research opportunity to use YOLOv8 and FaceNet for attendance systems.

III. MATERIALS AND METHOD

We developed a research workflow that begins with data and literature collection, followed by data preprocessing to prepare the LFW dataset in a suitable format for use. Next, two methods were implemented: YOLOv8-FaceNet and Dlib-ResNet, which were used to detect and recognize faces, respectively. After implementation, the performance of both methods was evaluated. This includes processing speed and recognition accuracy, as determined by the results of face pair testing. The goal is to compare both approaches and their effectiveness for a real-time face attendance system. Fig. 1 shows the research workflow.

A. IFDRS for Attendance System

Our proposed face recognition evaluation framework for attendance systems consists of two main stages: threshold calibration and face pair evaluation. This pipeline is implemented on two configurations: YOLOv8+FaceNet and Dlib-ResNet using the LFW dataset [17]. YOLOv8 is used for face detection (boundary box) while FaceNet and Dlib-ResNet are used for face recognition (identity embedding).

In achieving IFDRS, some systems combine face detection and recognition techniques. For example, YOLOv8 detects faces in video frames, and then FaceNet or Dlib-ResNet computes an embedding for each detected face. In such a pipeline, the detection speed directly affects the overall throughput, and the accuracy of the recognition model affects the final identification precision. Fig. 2 shows the architecture of our proposed attendance system.

B. YOLOv8+FaceNet for IFDRS

YOLOv8 is the eighth version of YOLO object detection, featuring an anchor-less detection head and a separate classification-regression branch for improved localization and accuracy [18]. The anchor-less architecture of YOLOv8 enhances its flexibility in handling a wide range of lighting environments. The model is designed to achieve high performance while maintaining real-time inference capabilities, especially in resource-constrained environments. In the context of face recognition systems, YOLOv8 serves as an efficient and accurate method for detecting facial regions under various lighting conditions, occlusions, and backgrounds.

The large Kernel Attention (LKA) module and the Bidirectional Feature Pyramid Network (BiFPN) integrated

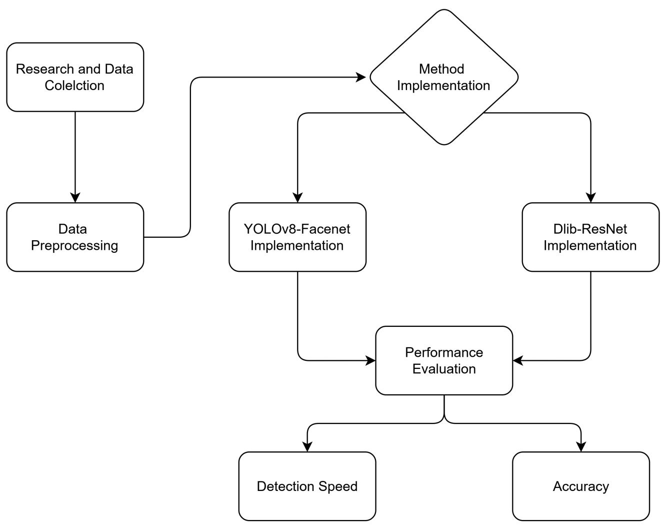

Fig. 1. The face recognition-based attendance research flow.

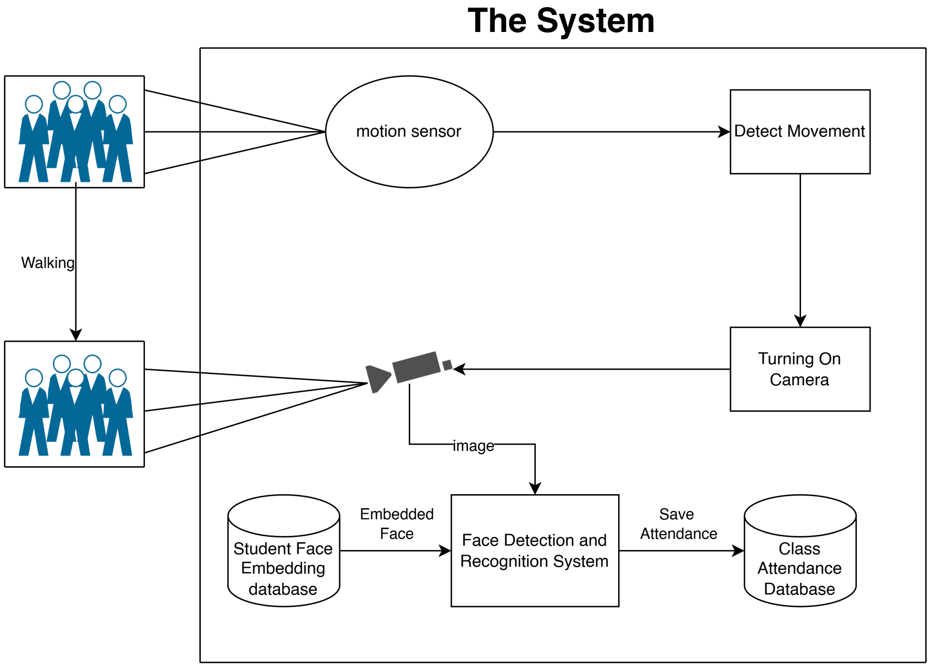

Fig. 2. Attendance system with face recognition architecture.

modified YOLOv8n model, which effectively makes it sensitive to subtle objects in dense visual scenes. LKA captures large spatial dependencies. It employs a depthwise convolution with a kernel of a specific size (symbolized as k). The following formula is used

$$\begin{aligned}\text{LKA}(x) = &\text{Conv}_{d=21}(x) + \\ &\text{Conv}_{d=5}(\text{Conv}_{d=7}(x))\end{aligned} \tag{1}$$

where Conv_d is a depthwise convolution with kernel size d, and x is the input feature map.

BiFPN in deep learning and computer vision is a multi-scale feature fusion method that has minimal overhead, thereby increasing efficiency in image detection [19]. The output feature map formula P_i from BiFPN, where I is the number of feature levels and J is the number of inputs, is as follows

$$P_i = \frac{\sum_j w_j \cdot P_j}{\sum_j w_j + \epsilon}, i \in I, j \in J \tag{2}$$

where P_j are input features, w_j are scalar weights, and ϵ is a constant to avoid division by zero.

In this study, YOLOv8 is used to obtain face bounding boxes from camera-captured images, reducing processing requirements before recognition. YOLOv8 is executed before the face recognition process is started, resulting in a much lower processing needed. Its low complexity and high Framepersecond (FPS) throughput make it a promising option for real-time class attendance monitoring when integrated with efficient recognition algorithms like Facenet or Dlib-ResNet [20].

FaceNet is a CNN that embeds facial images into a 128-dimensional feature space using a triplet loss function [21]. In practical systems, FaceNet is usually paired with a face detector (e.g., MTCNN) to crop and align faces before embedding. The triplet loss function used in FaceNet governs how the model embeds faces into the derivative feature space. The formula for the triple loss ($\mathcal{L}$) function is as follows:

$$\mathcal{L} = \sum_{i=1}^{N} [||f\{x_i^a\} - f\{x_i^p\}||_2^2|| - f\{x_i^a\} - f\{x_i^n\}||_2^2 + \alpha]_+ \tag{3}$$

where $f(x)$ is the output function of FaceNet, x_i^a is the anchor image, x_i^p is the positive image, x_i^n is the negative image, α is the margin parameter to increase separation, and $[z]_+$ is the $max(z, 0)$ function which is useful for eliminating violations.

We benchmark YOLOv8+FaceNet as IFDRS with Dlib+ResNet. Dlib+ResNet refers to the use of ResNet-34 by the Dlib library to generate 128D embeddings capable of performing face recognition [22]. Dlib's face recognition module is built on a variant of ResNet-34. Specifically, it utilizes a 34-layer CNN (with residual blocks) and incorporates a global average pooling layer to produce a 128-dimensional face embedding vector [23]. This 128-D output serves as a "facial fingerprint" for matching via simple Euclidean distance. Dlib's ResNet-34 has been fine-tuned on millions of faces (e.g., CASIA-WebFace) so that its embeddings are clustered based on identity [23].

C. Threshold Calibration and Test Metrics

Before performance evaluation, threshold calibration determines the best threshold to tell apart matched and mismatched face pairs. It works out the spread of similarity scores (cosine similarity), then graphs FAR and FRR agains different threshold values. The point where FAR and FRR are equal is called the Equal Error Rate (EER), and this threshold has been used in the final evaluation.

We propose an equal error rate (EER) calculation algorithm as shown in Algorithm 1. This algorithm uses the similarity scores of match and mismatch pairs. At each threshold value $\mathcal{T}$, the system calculates the number of false accepts (FA) and false rejects (FR). The determination of FA is based on false detections (F) and $\mathcal{T}$, while the determination of FR is based on true detections (T). The best variable $\mathcal{T}$ is the point where the difference between the false acceptance rate (FAR) and false rejection rate (FRR) is the smallest. Finally, the determination of EER is based on FAR and FRR.

Algorithm 1: EER Calculation Algorithm for Threshold Calibration.

Input: T, F, Steps
Output: EER, $\mathcal{T}_{opt}$

$thresholds \leftarrow$ linspace$(0.1, 1.0,$ Steps$)$;
FAR $\leftarrow$ new data structure;
FRR $\leftarrow$ new data structure;
for $i \leftarrow 0$ **to** *Steps* -1 **do**
  $\mathcal{T} \leftarrow thresholds[i]$;
  FA$\leftarrow$ calculate$(f \geq \mathcal{T},\ \forall\ f \in F)$;
  FR$\leftarrow$ calculate$(t < \mathcal{T},\ \forall\ t \in T)$;
  FAR$[i] \leftarrow \frac{\text{FA}}{|F|}$;
  FRR$[i] \leftarrow \frac{\text{FR}}{|T|}$;
end
$Diff \leftarrow |\text{FAR}[i] - \text{FRR}[i]|\ \forall\ i$;
$i_{min} \leftarrow \arg\min(Diff)$;
$\mathcal{T}_{opt} \leftarrow thresholds[i_{min}]$;
EER $\leftarrow \frac{\text{FAR}[i_{min}]+\text{FRR}[i_{min}]}{2}$;
return EER, $\mathcal{T}_{opt}$;

We use EER to optimize the threshold of face recognition. EER is the intersection between the FAR curve and the FRR curve [24]. The FAR curve typically decreases gradually as the threshold is increased. On the other hand, the FRR increases when the threshold is steadily increased. This method is crucial in biometric-based authentication because it explains the method's capability to distinguish genuine matches from impostors. The EER threshold curve can also show the characteristic of the curve near the EER point, where the slope indicates threshold robustness.

IV. Results and Discussion

A. Results

Before the evaluation, 500 matched and 500 mismatched pairs from the Labeled Faces in the Wild (LFW) dataset were prepared wherein preprocessing involved face detection with YOLOv8 and embedding extraction of FaceNet for all images. The procedure for testing is comprised of these two main phases which have been explained in the proposed system flow. The first phase deals with threshold calibration whereby the optimal threshold value is decided upon based on similarity distribution balancing the system between false acceptance and false rejection rates. In the second stage, an assessment is made of how well the face recognition system actually performs regarding accuracy, error metrics, and processing time. Model configurations tested in this exercise were YOLOv8+FaceNet and Dlib-ResNet. All tests were run on the LFW dataset comprising matched and mismatched identity pairs.

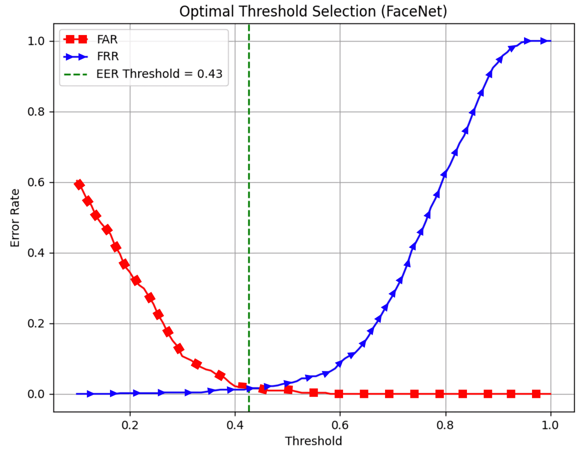

Fig. 3. FaceNet optimal threshold.

The facial recognition system is evaluated on 1,000 image pairs with a calibrated threshold. The image pairs include 500 matched pairs and 500 unmatched pairs. Threshold calibration also identifies the point where FAR and FRR are equal, i.e., EER. In other words, it ensures that the threshold used will adequately make a trade-off between accepting genuine users and rejecting intruders.

The first test involves optimizing the face recognition threshold on FaceNet using Equal Error Rate (EER). Fig. 3 shows that the optimal threshold for the YOLOv8+FaceNet configuration is found at 0.4273 with an EER of 0.0130. The stability of the EER indicates robustness, as minor threshold variations have negligible effects on recognition. The threshold at FAR 0.01 is 0.4273, and the FRR at that value is 0.012. This demonstrates that this authentication can be utilized in systems such as student attendance, as student attendance requires a low FAR with high FRR tolerance.

The next test is to optimize the threshold on the Dlib-ResNet method. Fig. 4 shows the EER curve on Dlib-ResNet. For Dlib-ResNet, the optimal threshold is 0.3909 with an EER of 0.0100. The change in the EER of the Dlib-ResNet model is greater than that of YOLOv8-FaceNet, indicating that the EER of this model is less robust, or that small threshold changes have a more significant impact on face recognition. The threshold at FAR 0.01 is 0.3909, and the FRR at that value is 0.016. This makes it less suitable for student attendance compared to YOLOv8+FaceNet, as it results in a higher FAR and FRR.

Finally, we compare the performance of the model using the threshold search method in terms of the EER and with the default threshold of 0.5. Fig. 5 shows a comparison of five face recognition performance metrics: accuracy, precision, recall, F1-score, and area under the curve (AUC). The performance improvement with EER-based threshold optimization is proven to increase accuracy and recall. Only the precision value of the model decreased, which means this threshold is more tolerant of impostors. However, the F1-score indicates that the overall performance of the optimum threshold remains superior. A higher AUC shows that this threshold has a better balance between the true positive rate (TPR) and false positive

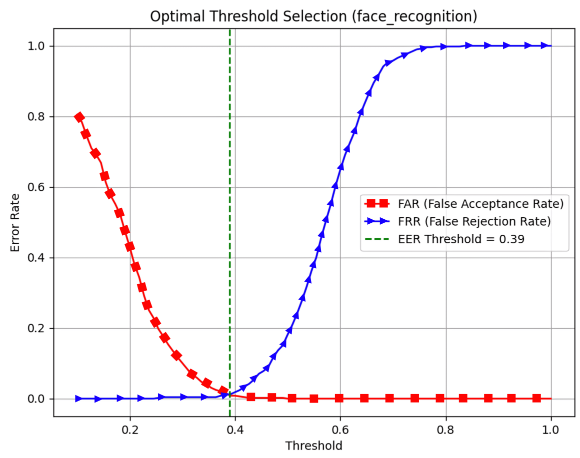

Fig. 4. Dlib-ResNet optimal threshold.

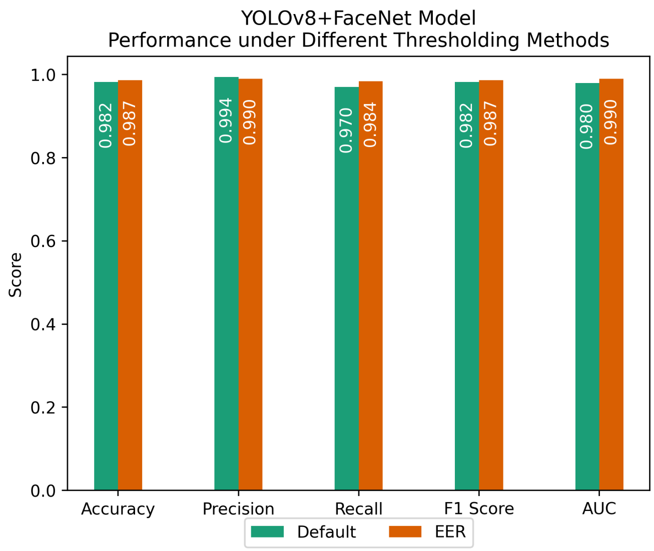

Fig. 5. Comparison of face recognition performance based on thresholds: default and EER optimized.

rate (FPR).

B. Discussion

The test results show an EER of 0.013 for YOLOv8+FaceNet and an EER of 0.010 for Dlib-ResNet. Rahman *et al.* assert that EER with this range is classified as low. The study also emphasizes that the EER remains flat with minimal changes, indicating that the model is suitable for environments with varying lighting, as changes in threshold at low EER do not result in significant performance changes in FAR and FRR.

Dixit *et al.* [25] proposed an attendance system using face recognition, where the method behind it is Dlib-ResNet face recognition. However, the accuracy aspect of the technique can still be improved. On the other hand, Sunitha *et al.* [26] said that the YOLOv8 method has better accuracy than other models because of the single-pass processing, which increases detection speed and processing. Table II shows that our proposed YOLOv8+FaceNet method has better accuracy than the state-of-the-art, which is 98.7% compared to 98.6%. The

TABLE II
IFDRS Performance Comparison for Student Attendance.

Model	**Accuracy**	**FAR**	**FRR**	**EER**
Dlib-ResNet [25]	98.60%	0.0080	0.0120	0.0100
YOLOv8+FaceNet	**98.70%**	**0.0100**	**0.0160**	**0.0130**

contribution of our research is twofold. First, a novel IFDRS method, YOLOv8+FaceNet, is presented, which achieves higher accuracy than the state-of-the-art method. Second, we propose an IFDRS method with optimum accuracy as a face recognition-based attendance system.

Several studies have employed EER to determine the optimal threshold in face recognition. Bae *et al.* [27] found that expression influences in determining EER. The study stated that a normal face has an EER of 0.065% while an expressive face has an EER of 1.13%. On the other hand, Chou *et al.* [28] stated that the use of a fixed threshold in FaceNet is because the method is sensitive to environmental conditions such as lighting levels. There has never been a study that uses EER to determine the optimum threshold in FaceNet. Our analysis then shows that the optimum threshold for FaceNet is 0.43, with an EER value of 0.0013. The contribution of our study is a method for determining the optimal threshold for FaceNet using EER.

V. Conclusion

This study proposed and evaluated a real-time attendance system based on face recognition by comparing two model configurations: YOLOv8+FaceNet and Dlib-ResNet. The system testing utilized the LFW dataset, employing an evaluation approach based on matched and mismatched face pairs. The two main stages in the testing include threshold calibration based on EER and system performance evaluation based on accuracy, recognition error, and processing time. The test results indicate that the EER of the YOLOv8+FaceNet model is 0.013, with FAR and FRR of 0.010 and 0.016, respectively. These values are at a threshold of 0.4273. Then, the EER of the Dlib+ResNet model is 0.0010, with FAR and FRR of 0.008 and 0.012. The optimum threshold of the model is 0.3909. Performance testing shows that the proposed YOLOv8+FaceNet model achieves slightly higher accuracy than Dlib-ResNet (98.7% vs. 98.6%). Future work should evaluate inference speed across models, robustness under challenging conditions (e.g., shadows, occlusions), and feasibility on resource-constrained devices. Additionally, upcoming work could investigate privacy-aware attendance architectures that balance recognition performance with data protection.

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